WATER QUALITY ANALYSIS

Name: Divya Jha

Registration no. 12417664

Programme and Section. B.Tech. In CSE, K2M030

Lovely Professional University

Phagwara, Punjab 144411

School of Computer Science and Engineering

Abstract- This paper presents a data analysis of the BKB Water Quality Dataset (1989-2019). Six water quality measurements were studied: pH, Salinity, Dissolved Oxygen (DO), Water Temperature, Water Depth, and Secchi Depth. The analysis is conducted using Python-based data science tools, including Pandas for data manipulation, and Matplotlib and Seaborn for visualization. The methodology involves data preprocessing, exploratory data analysis (EDA), correlation analysis, and predictive modelling using linear regression.

Key analyses provides some important observations, such as the correlation between temperature and dissolved oxygen levels, the presence of outliers, most water samples have a normal pH (Normal class), and a few unusual readings exist at the extremes and the general trends regarding water quality changes. The study highlights the potential of data-driven methodologies in deriving actionable insights from large datasets.

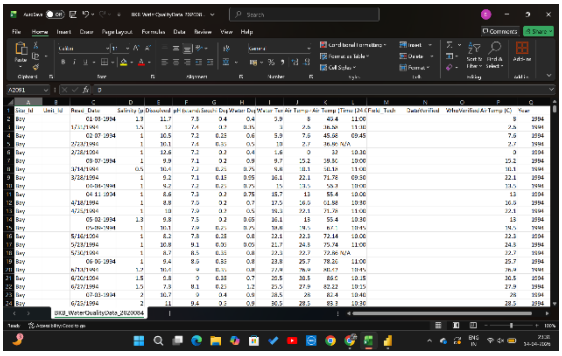
I. INTRODUCTION

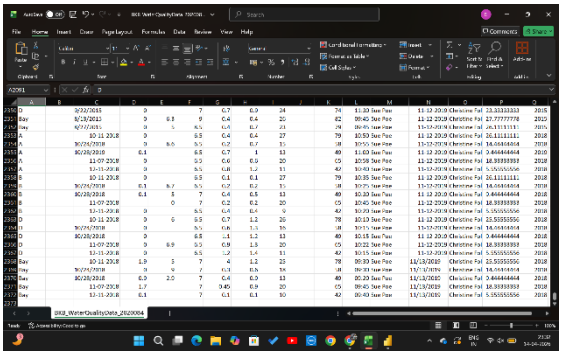
The Water Quality data set contains complete information on many environmental factors that need to be considered while examining the status of any body of water. The Water Quality data set contains details such as water temperature, oxygen concentration, pH level, and salinity. These environmental factors have great importance in determining water quality.

With the help of data science and visualization techniques, such environmental datasets can be effectively analyzed to identify patterns, relationships, and trends. This project, titled **"Water Quality Analysis,"** focuses on examining the behavior of key water quality parameters using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn.

Some of the analysis techniques employed by the study include the analysis of the distribution of essential variables, the correlation between parameters like temperature and dissolved oxygen, and the analysis of trends. Some of the visualization methods utilized in this research include histogram, scatter plots, boxplot, heatmap, and line graph. In addition, a linear regression model was employed for predicting the amount of dissolved oxygen in the water and assessed using the mean squared error (MSE).

II. Source of Dataset





1.The dataset used in this project is sourced from the "The Home of the U.S. Government's Open Data".

2.The specific dataset analyzed is titled "BKB_WaterQualityData_2020084", available at: https://catalog.data.gov/dataset/water-quality-data

3.This dataset provides detailed environmental and water quality information that were collected during repeated observations. The dataset includes important variables like water temperature, dissolved oxygen, pH value, and salinity, which are crucial to determine the status of the water bodies.

4.Key features of the dataset include:

1. pH (standard units)
2. Salinity (ppt)
3. Dissolved Oxygen (mg/L)
4. Water Temp (C)
5. Water Depth (m)
6. Secchi Depth (m)

5.The dataset is provided in a structured tabular format, suitable for analysis using tools like Python, Excel, or R.

6.The BKB_WaterQualityData_2020084 data was collected by the U.S. Fish and Wildlife Service and is comprehensive datasets for understanding water quality and the impact of environmental factors on it.

7.The dataset is open-source and available for public use under the U.S. Government's Open Data, enabling researchers and students to conduct in-depth analysis for academic and policy-related purposes.

III. EDA process

The first step in the exploratory data analysis was to assess the completeness of the dataset by checking for missing or null values. This was crucial to ensure data integrity before proceeding with analysis or visualization.

A. Process:

- The dataset was loaded into a Pandas Data Frame using the read_csv() function.
- The isnull().sum() function was applied to identify the number of null values in each column.

B. Findings:

- Most of the key columns such as pH, Salinity , Dissolved Oxygen, Water temp(c), Water Depth(m) and Secchi Depth(m) were found to have null values.
- A few less relevant columns also had missing data, but these were not used in the core analysis and visualizations.

C. Action Taken:

- For the essential columns with nulls, either:
  - The rows with null data were removed as they could impact on the analysis.
  - Also the columns were renamed for simplicity].
- For any non-essential columns with nulls, either:
  - The rows were retained as they did not impact the outcome.
  - Or the columns were excluded from specific analyses where complete data was required.

D. Conclusion:

The dataset had missing values in many columns, which were handled using appropriate data cleaning techniques to ensure smooth and accurate analysis.

```
       'Dissolved Oxygen (mg/L)', 'pH (standard units)', 'Secchi Depth (m)',
       'Water Depth (m)', 'Water Temp (?C)', 'Air Temp-Celsius',
       'Air Temp (?F)', 'Time (24:00)', 'Field_Tech', 'DateVerified',
       'WhoVerified', 'AirTemp (C)', 'Year'],
      dtype='object')
Duplicate rows: 0
Site_Id                 1
Read_Date               5
Salinity              130
DissolvedOxygen       851
pH                     95
Secchi_Depth           73
Water_Depth            71
Water_Temp            121
Air Temp-Celsius     2286
Time (24:00)           63
Field_Tech             39
Year                    0
dtype: int64
Site_Id              0
Read_Date            0
Salinity             0
DissolvedOxygen      0
pH                   0
Secchi_Depth         0
Water_Depth          0
Water_Temp           0
Air Temp-Celsius     0
```

IV. Dataset Structure and Summary Statistics

After ensuring that the dataset was largely complete and free from missing values, the next step was to understand the structure and summary statistics of the data. This helps provide an overview of the types of data present, their distributions, and any anomalies.

A. Dataset Overview with .info()

The .info() function was used to:

1. Display the total number of entries and columns.

2. Show data types (e.g., int64, object) for each column.

3. Identify the memory usage and whether any columns contain null values.

Findings:

URL: https://catalog.data.gov/dataset/water-quality-data

1. The dataset consisted of both numerical and categorical columns.

2. Important attributes pH, Salinity , Dissolved Oxygen, Water temp(c), Water Depth(m) and Secchi Depth(m) were all of numeric type (float64), making them suitable for quantitative analysis.

3. Categorical data like Site_Id and Field_Tech were of type object.

```
None
       Salinity (ppt)  Dissolved Oxygen (mg/L)  ...  AirTemp (C)         Year
count     2241.000000              1520.000000  ...  2371.000000  2371.000000
mean         0.717068                 6.646263  ...    15.663050  2006.049346
std          1.230819                 2.506608  ...    10.314720     9.002373
min          0.000000                 0.000000  ...   -17.777778  1899.000000
25%          0.000000                 4.800000  ...     8.888889  2001.000000
50%          0.000000                 6.500000  ...    16.666667  2007.000000
75%          1.000000                 8.500000  ...    23.888889  2012.000000
max          9.000000                15.100000  ...    33.500000  2019.000000
```

B. Statistical Summary with .describe()

The .describe() function provided the following for each numerical column:

1. Count – Total number of non-null values

2. Mean – Average value

3. Standard Deviation (std) – Measure of variation

4. Min / Max – Range of values

5. 25%, 50%, 75% Quartiles – Spread of the data

Findings:

1.Parameters like dissolved oxygen, air temperature, and salinity appear to vary significantly, which may suggest changing conditions from one observation to another.

2. Some variables had extreme values (high max compared to the median), indicating skewed distributions.

3. The above observations show that visualizations (e.g., histogram, box plots, heat maps) may be crucial in making sense of the results of any statistical analyses.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2371 entries, 0 to 2370
Data columns (total 17 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Site_Id                  2370 non-null   object
 1   Unit_Id                  32 non-null     object
 2   Read_Date                2366 non-null   object
 3   Salinity (ppt)           2241 non-null   float64
 4   Dissolved Oxygen (mg/L)  1520 non-null   float64
 5   pH (standard units)      2276 non-null   float64
 6   Secchi Depth (m)         2298 non-null   float64
 7   Water Depth (m)          2300 non-null   float64
 8   Water Temp (?C)          2250 non-null   float64
 9   Air Temp-Celsius         85 non-null     float64
 10  Air Temp (?F)            2300 non-null   float64
 11  Time (24:00)             2308 non-null   object
 12  Field_Tech               2332 non-null   object
 13  DateVerified             453 non-null    object
 14  WhoVerified              453 non-null    object
 15  AirTemp (C)              2371 non-null   float64
 16  Year                     2371 non-null   int64
dtypes: float64(9), int64(1), object(7)
memory usage: 315.0+ KB
None
```

Conclusion:

Using .info() and .describe(), the dataset was confirmed to be:

1. Numerically rich and structurally suitable for analysis.

2. Containing wide variations across different parameters, reflecting changing environmental conditions and possible fluctuations in water quality over time.

Objective 1: To analyze the distribution of data using histogram plots.

i. Introduction

This analysis aims to understand the distribution of key water quality parameters such as pH,salinity and dissolved oxygen.

ii. General Description

The data set comprises a number of environmental variables, which include water temperature, dissolved oxygen, pH, and salinity. In this case, histograms will be used to illustrate the frequency distribution of certain variable values (such as pH , salinity and dissolved oxygen).

iii. Specific Requirements, Functions, and Formulas

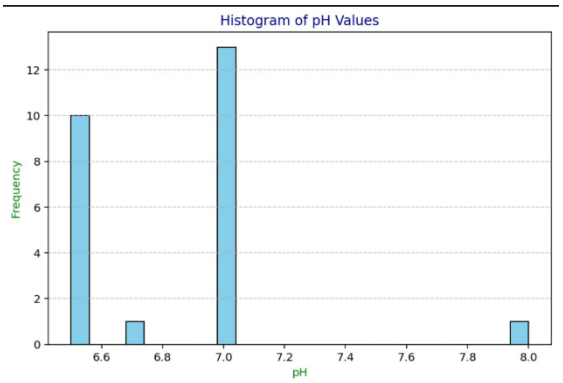
- Data Selection: Selecting numerical columns such as pH, salinity or DissolvedOxygen

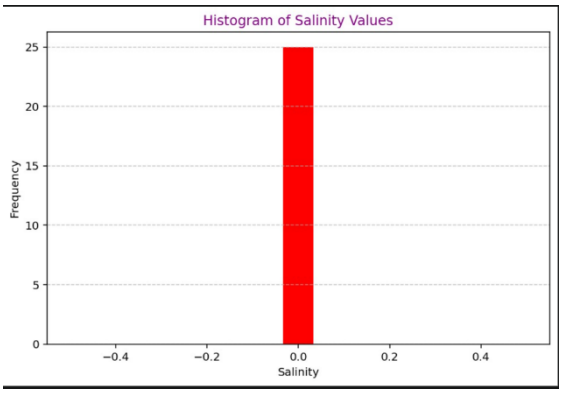
- Visualization: matplotlib.pyplot.hist() – used to create histogram plots.

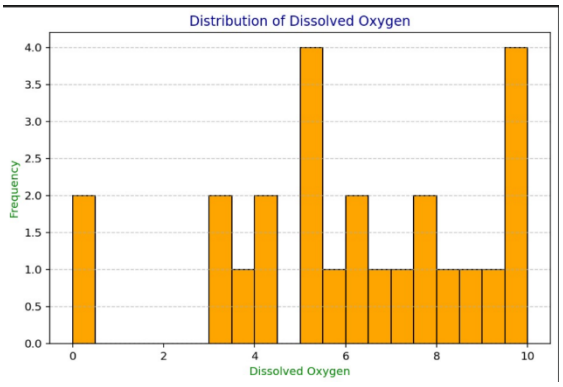
- Bins: Defines the number of intervals for grouping data values.

Code Snippet:

```
#OBJECTIVE 1: To analyze the distribution of data using histogram plots.
#Histogram of pH
plt.figure(figsize=(8,5))
plt.hist(data["pH"], bins=25, color='skyblue', edgecolor='black')
plt.title("Histogram of pH Values", color='darkblue')
plt.xlabel("pH", color='green')
plt.ylabel("Frequency", color='green')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()

#Histogram of salinity
plt.figure(figsize=(8,5))
plt.hist(data["Salinity"], bins=15,color='red')
plt.title("Histogram of Salinity Values" ,color='purple')
plt.xlabel("Salinity", color='black')
plt.ylabel("Frequency", color='black')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()

#Histogram of DissolvedOxygen
plt.figure(figsize=(8,5))
plt.hist(data["DissolvedOxygen"], bins=20, color='orange' , edgecolor='black')
plt.title("Distribution of Dissolved Oxygen", color='darkblue')
plt.xlabel("Dissolved Oxygen", color='green')
plt.ylabel("Frequency", color='green')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```







iv. Analysis Results

After processing the dataset:

- There is no variation in salinity (it's constant) - Salinity histogram depicts all the data in just one place (around 0). There is no variation in salinity among the provided values, which either implies stability or problems with measurement of these values.
- There is very low variation in the pH level, and it is neutral - All the pH data is located close to value of 7, and it depicts a situation where pH is constant, meaning that water is neutral (neither acidic nor basic).
- The Dissolved oxygen level is highly variable - The oxygen level graph spreads over a wide range, lying between 0 and 10. This indicates that the oxygen level is highly variable and fluctuates considerably.

v. Visualization

The histograms show that salinity remains nearly constant across all observations, indicating little to no variation in that parameter. In contrast, pH values are tightly clustered around neutral (approximately 7), suggesting stable and consistent water conditions. However, dissolved oxygen levels display a wide distribution, ranging from low to high values, which indicates significant variability likely influenced by environmental factors such as temperature, biological activity, or water movement.

Objective 2: To Analyze Relationship Between pH and Dissolved Oxygen

i. Introduction

The study seeks to analyze how pH and dissolved oxygen affect each other. It is necessary to investigate their relationship because they play a crucial role in assessing the quality of water.

ii. General Description

The dataset contains separate columns for environmental attributes like pH value and dissolved oxygen . For this objective, these two variables are analyzed together to determine whether any relationship or pattern exists between them.

iii. Specific Requirements, Functions, and Formulas

- Data Selection: Selecting pH and Dissolved Oxygen columns
- Scatter Plot: plt.scatter() – used to visualize the relationship between two variables.

Code Snippet:

```
#OBJECTIVE 2: To Analyze Relationship Between pH and Dissolved Oxygen
plt.figure(figsize=(8,5))
plt.scatter(data["pH"], data["DissolvedOxygen"], color='red')
plt.title("pH vs Dissolved Oxygen", color='green')
plt.xlabel("pH", color='blue')
plt.ylabel("Dissolved Oxygen", color='blue')
plt.show()
# the scatter plots shows that the points are loosely packed and are in one direction
# the correlation among them is weak.
```

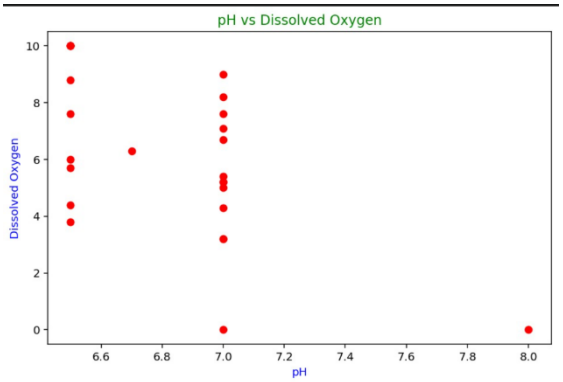
iv. Analysis Results

The analysis compares male and female-headed households across different tehsils:

- There is no strong linear correlation between pH and dissolved oxygen (DO). Values are widely scattered at each pH level.
- Most of the data is around pH 7, so the samples are mainly neutral .
- There is no clear trend—oxygen does not steadily increase or decrease with pH .
- At the same pH (especially 7), dissolved oxygen changes a lot, so pH alone doesn't control it .
- There are unusual points (outliers) like zero oxygen at pH 7 and 8.

v. Visualization

The scatter plot shows how pH relates to dissolved oxygen levels in the samples. Most data points cluster around neutral pH (≈7). At this level, dissolved oxygen values vary widely. There is no clear trend showing that dissolved oxygen increases or decreases with pH. A few outliers exist, including very low or zero oxygen values. Overall, the visualization indicates a weak relationship between pH and dissolved oxygen.



Objective 3: Implementation of Linear Regression for Predictive Analysis among water temp and dissolved oxygen

i. Introduction

The purpose of this analysis is to analyze the relationship between the temperature of water and the dissolved oxygen content of the water using a predictive model. The goal is to find out the effect of temperature on dissolved oxygen content and develop a predictive model for the same.

ii. General Description

The dataset includes environmental parameters such as water temperature and dissolved oxygen. For this objective, water temperature is treated as the independent variable, while dissolved oxygen is the dependent variable. Linear Regression is applied to model their relationship and make predictions..

iii. Specific Requirements, Functions, and Formulas

• Data Selection: Selecting Water_Temp (independent variable) and DissolvedOxygen (dependent variable)

• Model Creation: LinearRegression() – used to build the regression model

• Model Training: .fit(X, y) – trains the model using input data

• Prediction: .predict(X) – generates predicted values

• Evaluation: mean_squared_error() – calculates prediction error

• Visualization: matplotlib.pyplot.scatter() and plt.plot() – used to display actual data and regression line

Code Snippet:

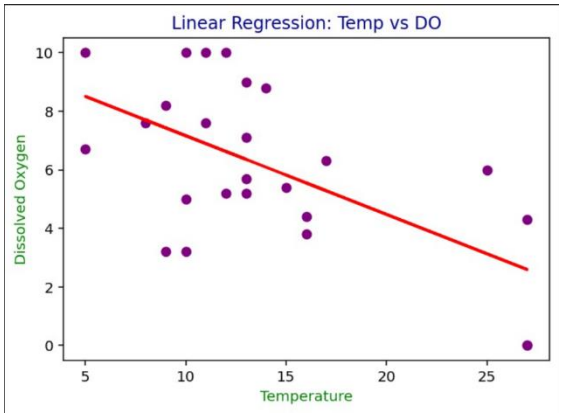
```
# Prepare data
X = data[["Water_Temp"]].dropna()
y = data.loc[X.index, "DissolvedOxygen"]

# Train model
model = LinearRegression()
model.fit(X, y)

# Predict
y_pred = model.predict(X)

# Calculate MSE
mse = mean_squared_error(y, y_pred)
print("Mean Squared Error:", mse)

# Plot
plt.scatter(X, y, color='purple')   # scatter color
plt.plot(X, y_pred, color='red', linewidth=2)   # regression line with width
plt.title("Linear Regression: Temp vs DO", color='darkblue')
plt.xlabel("Temperature", color='green')
plt.ylabel("Dissolved Oxygen", color='green')
plt.show()
```

iv. Analysis Results

- Negative relationship: The regression line slopes downward, showing that dissolved oxygen decreases as temperature increases.
- Moderate scatter: Data points are somewhat spread out around the regression line, indicating the relationship exists but is not perfectly strong.
- Scientific relevance: This pattern aligns with the known principle that warmer water holds less dissolved oxygen.

v. Visualization

From the graph, it can be seen that the scatter plot is accompanied by the fitted linear regression equation. It can be observed from the negative slope of the linear regression equation that there is a negative relationship, whereby the higher the value of the variable "Temperature", the lower the value of the variable "Dissolved Oxygen". It is evident that the higher the value of the variable "Dissolved Oxygen", the lower the value of the variable "Temperature".



Objective 4. Year-wise Trend Analysis

i. Introduction

The focus here is on examining how much dissolved oxygen levels fluctuate from year to year. This is important because studying the trend for every year gives us an idea about the changing water quality and environment.

ii. General Description

The dataset contains water quality parameters recorded across different years. For this objective, the Year column is used to group the data, and the average dissolved oxygen values are calculated for each year. This helps in analyzing how oxygen levels have changed over time.

iii. Specific Requirements, Functions and Formulas

- Grouping: groupby("Year") – groups data based on year
- Aggregation: .mean() – calculates average dissolved oxygen per year
- Visualization: matplotlib.pyplot.plot() – used to create a line graph
- Data Check: value_counts() – used to verify number of entries per year

Code Snippet:

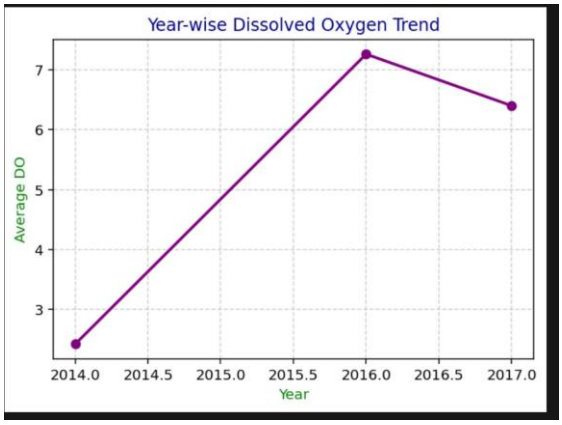
```
#OBJECTIVE 4: Year-wise Trend Analysis
year_analysis = data.groupby("Year")["DissolvedOxygen"].mean()
plt.plot(year_analysis.index, year_analysis.values,
         color='purple', linewidth=2, marker='o')
plt.title("Year-wise Dissolved Oxygen Trend", color='darkblue')
plt.xlabel("Year", color='green')
plt.ylabel("Average DO", color='green')
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()
#the graph is this way because their is no entry of the year 2015
data["Year"].value_counts() #tells me how many entries per year
```

iv. Analysis Results

- Dissolved oxygen shows a strong increase from 2014 to 2016, indicating significant improvement over this period.
- The decrease in 2017 is not drastic, indicating relatively stable oxygen levels after 2016.
- Changes over the years suggest that external factors (e.g., climate, pollution, or management practices) may have affected dissolved oxygen levels.

v. Visualization

The line graph provides information regarding the annual variation in the level of dissolved oxygen from the year 2014 to 2017. There is an apparent rise in the level of dissolved oxygen from the year 2014 to 2016, after which there is a dip in the year 2017. The highest level of dissolved oxygen is reached in the year 2016.



Objective 5. Outlier Detection using Boxplot for Dissolved oxygen and pH

i. Introduction

This analysis focuses on identifying outliers in key water quality parameters such as dissolved oxygen and pH. Detecting outliers is important as they may indicate abnormal environmental conditions, measurement errors, or significant changes in water quality.

ii. General Description

The dataset contains numerical environmental parameters including dissolved oxygen and pH values. For this objective, boxplots are used to visualize the distribution of these variables and detect any extreme values that lie outside the normal range.

iii. Specific Requirements, Functions and Formulas

1. Libraries: pandas, seaborn, matplotlib

2. Dataset columns: DissolvedOxygen, pH

3. **Visualization:** matplotlib.pyplot.boxplot() – used to create boxplots

Code Snippet:

```
#OBJECTIVE 5: Outlier Detection using Boxplot
# pH Boxplot
plt.boxplot(data["pH"].dropna(),
            patch_artist=True,
            boxprops=dict(facecolor='lightgreen', linewidth=4),
            whiskerprops=dict(color='purple', linewidth=4),
            capprops=dict(color='green', linewidth=4),
            medianprops=dict(color='red', linewidth=4))
plt.title("Boxplot of pH", color='darkblue')
plt.ylabel("pH", color='green')
plt.show()

# Dissolved Oxygen Boxplot
plt.boxplot(data["DissolvedOxygen"].dropna(),
            patch_artist=True,
            boxprops=dict(facecolor='lightblue', linewidth=4),
            whiskerprops=dict(color='green', linewidth=4),
            capprops=dict(color='blue', linewidth=4),
            medianprops=dict(color='red', linewidth=4))
plt.title("Boxplot of Dissolved Oxygen", color='darkblue')
plt.ylabel("Dissolved Oxygen", color='green')
plt.show()
```

iv. Analysis Results

1.Dissolved oxygen exhibits high variability with a broad range (approximately 0–10 mg/L). On the other hand, pH values cluster around neutral conditions.

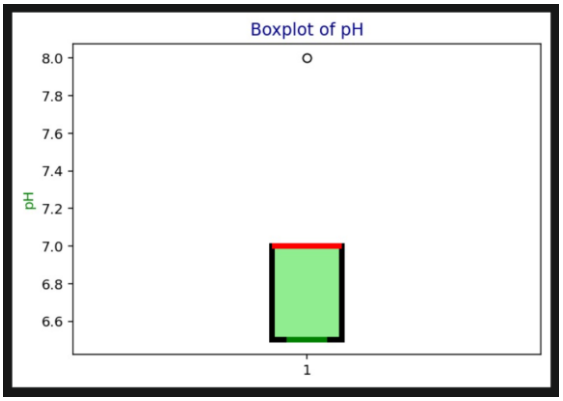
2. The median Dissolved Oxygen (~6 mg/L) indicates moderate oxygen levels, whereas the median pH (~7) confirms generally neutral water conditions.

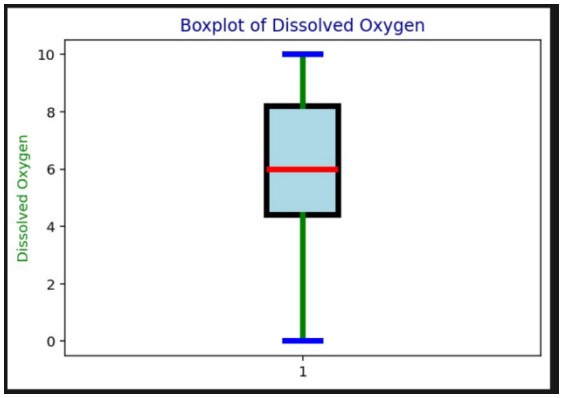
3. Dissolved Oxygen has a higher IQR (≈4.5-8 mg/l), which indicates variability in environmental conditions. In contrast, pH has a very low IQR (≈6.5-7.0).

4. No significant outliers in Dissolved Oxygen, but pH has one outlier (~8), suggesting an occasional increase in alkalinity.

v. Visualization

The boxplots have been utilized for representing the distribution of Dissolved Oxygen and pH data. It can be seen from the graph that there is more variation in Dissolved Oxygen as compared to pH. In addition to this, the pH graph represents the data values around neutral with a narrower range than Dissolved Oxygen, which is also evident from the existence of an outlier in the pH data.





Objective 6: Correlation Analysis Between Parameters

i. Introduction

The objective of this study is to determine the correlations among various water quality variables like temperature, dissolved oxygen, pH, and salinity. It is essential to know these relationships to assess the impact that one variable has on another.

ii. General Description

The dataset contains multiple numerical environmental parameters related to water quality. For this objective, a correlation matrix is computed to measure the strength and direction of relationships between variables. A heatmap is used to visually represent these correlations, making it easier to identify strong positive or negative relationships among the parameters.

iii. Specific Requirements, Functions, and Formulas

- Dataset columns: numerical columns such as Water_Temp, DissolvedOxygen, pH, Salinity, etc.
- **CorrelationMatrix:** .corr(numeric_only=True) – calculates correlation between numerical variables
- **Visualization:** matplotlib.pyplot.imshow() – used to create heatmap

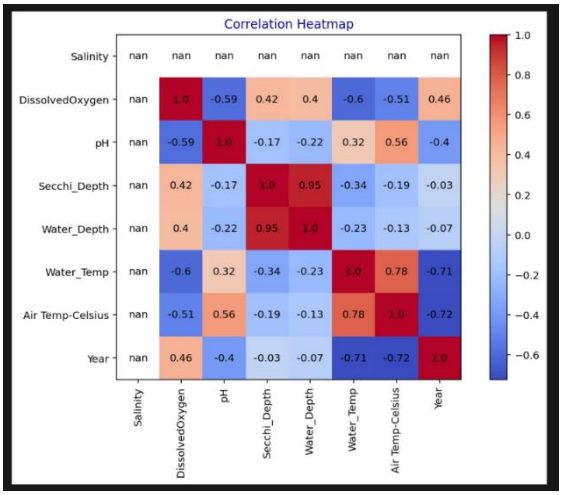
Code Snippet:

```
#OBJECTIVE 6: Correlation Analysis Between Parameters
# Correlation matrix
corr = data.corr(numeric_only=True)
plt.figure(figsize=(10,6))

# Heatmap
plt.imshow(corr, cmap='coolwarm')
plt.colorbar()

# Add values inside boxes
for i in range(len(corr.columns)):
    for j in range(len(corr.columns)):
        plt.text(j, i, round(corr.iloc[i, j], 2),
                 ha='center', va='center', color='black')

# Labels
plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)
plt.yticks(range(len(corr.columns)), corr.columns)
plt.title("Correlation Heatmap", color='darkblue')
plt.show()
```

iv. Analysis Results

- Strong positive relationships: Secchi depth and water depth show a very strong positive correlation (≈ 0.95), indicating clearer water is associated with greater depth.
- Temperature linkage: Water temperature and air temperature are highly positively correlated (≈ 0.78), suggesting atmospheric conditions strongly influence water temperature.
- Dissolved oxygen patterns: Dissolved oxygen has a moderate negative correlation with water temperature (≈ -0.60) and air temperature (≈ -0.51), implying oxygen levels decrease as temperature rises.
- Temporal trends: Year shows a strong negative correlation with both water temperature (≈ -0.71) and air temperature (≈ -0.72), indicating a decreasing trend in temperatures over time in this dataset.

v. Visualization



Objective 7: Water Quality Classification

i. Introduction

The main point of the analysis is classification of the water quality depending on the pH value into groups of acid, neutral, and alkali waters. Classification is essential when analyzing the quality of water because the pH is crucial in establishing its quality.

ii. General Description

The dataset contains pH values for different water samples. For this objective, a classification approach is used where each sample is categorized based on its pH level. A new column named Water_Quality is created to store these categories. The distribution of these categories is then visualized using a bar chart to understand the proportion of different water quality types.

iii. Specific Requirements, Functions, and Formulas

• Dataset column: pH

• **Custom Function:** Defines rules to classify water into Acidic, Normal, and Alkaline

• **Apply Function:** .apply() – used to apply classification to each pH value

• **Visualization:** plot(kind="bar") – used to display category distribution

Code Snippet:

```
#OBJECTIVE 7: Water Quality Classification
def water_quality(ph):
    if ph < 6.5:
        return "Acidic"
    elif ph <= 8.5:
        return "Normal"
    else:
        return "Alkaline"

data["Water_Quality"] = data["pH"].apply(water_quality)

# Count categories
data["Water_Quality"].value_counts().plot(kind="bar")
plt.title("Water Quality Categories")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()
```

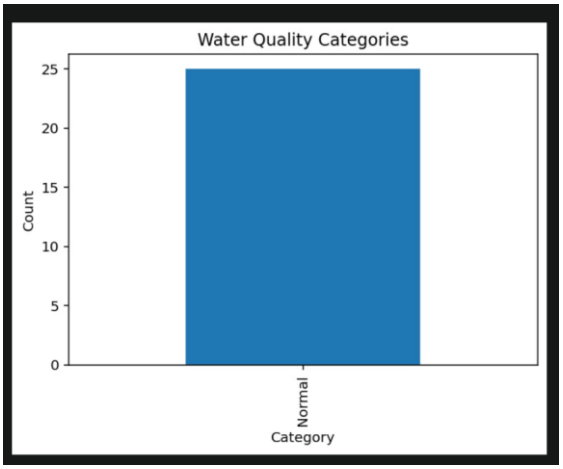
iv. Analysis Results

1. All samples fall under the "Normal" category, showing no variation in water quality.

2. No instances of poor or contaminated water were observed in the dataset.

3. The lack of category diversity limits deeper comparative analysis.

v. Visualization



V. Conclusion

The project titled **"Water Quality Analysis and Prediction using Python"** demonstrates the effective application of data science techniques to analyze environmental datasets. Using a water quality dataset, this study provided a detailed exploration of key parameters such as water temperature, dissolved oxygen, pH, and salinity, helping to uncover important patterns, relationships, and variations in water conditions.

With the help of libraries like pandas, numpy, matplotlib, and seaborn , the importance of Python is shown in dealing with practical environmental data sets. This was followed by data pre-processing, where missing values were handled, and descriptive statistics were checked to ensure the accuracy of the data set.

Various visualization techniques such as histograms, scatter plots, boxplots, heatmaps, line graphs, and bar charts were used to understand data distribution, detect outliers, and analyze trends over time. The heatmap revealed significant correlations between parameters, particularly the inverse relationship between water temperature and dissolved oxygen, which aligns with scientific understanding.

Moreover, Linear Regression allowed for predictive analysis to occur, whereby one could demonstrate how one could estimate dissolved oxygen using water temperature. Through the evaluation process, which included measures like Mean Squared Error (MSE), it was possible to measure the accuracy and effectiveness of the model, thereby lending validity to the results.

In conclusion, this project has not only improved our skills in data analysis and machine learning but has also contributed to important knowledge about the environmental status and the dynamics of water quality. This project is an example of how important it is to rely on data when dealing with environmental issues.

VI. Future Scope

As issues related to the environment keep increasing and the use of data becomes more important for making decisions, there is a lot of room left in this water quality analysis research project. The current study provides a strong foundation that can be extended into more advanced and impactful applications.

Another feature that could be included for development in the future is the analysis based on Time Series. Using information from multiple years' worth of readings or even constant monitoring, we could examine trends in water quality factors like pH levels, temperature, and dissolved oxygen over a longer period of time.

Another important extension is the use of Geospatial Analysis. By integrating tools such as GeoPandas or GIS platforms, water quality data can be visualized on maps to identify region-specific patterns and pollution hotspots. This would enable better monitoring and targeted environmental management.

Future work can also focus on **Multi-Parameter Analysis**, combining more environmental and chemical indicators to gain deeper insights into water quality and ecosystem health. This would improve the accuracy and reliability of analysis.

Finally, the creation of an interactive dashboard with software such as Plotly, Dash, or Tableau could be very useful. This would enable users to interact with the data, find trends within it, and keep track of their water quality measurements.

VII. References

1. "BKB_WaterQualityData_2020084", available at: https://catalog.data.gov/dataset/water-quality-data

2. **United States Geological Survey.** *Water Quality Data and Monitoring.* https://www.usgs.gov/mission-areas/water-resources

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4. **World Health Organization.** *Guidelines for Drinking-water Quality.* https://www.who.int/publications/i/item/9789241549950

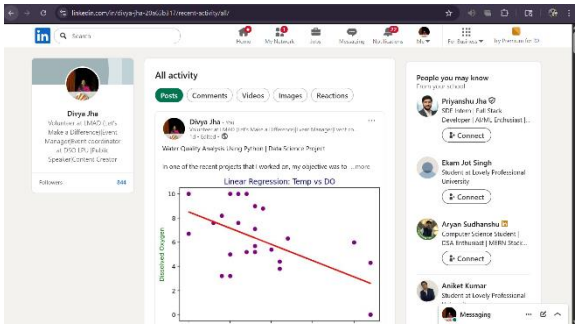
5. Hunter, J.D. (2007). *Matplotlib: A 2D Graphics Environment*. Computing in Science & Engineering.

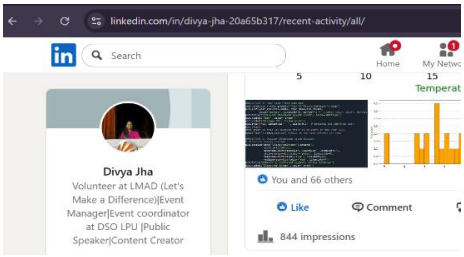
6. Jake VanderPlas (2016). Python Data Science Handbook.

7. Data.Gov.usa , An official website of Government of United States

Lovely Professional University. *INT375 Python Programming Project Guidelines.*

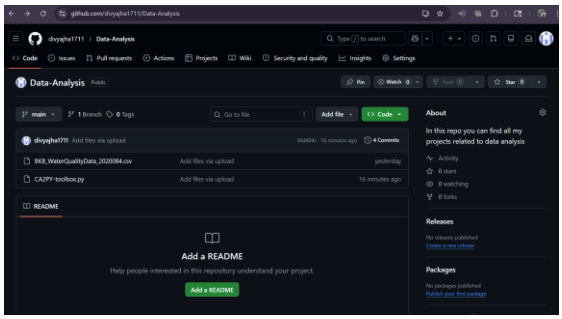
VIII. LinkedIn Post





Link : https://www.linkedin.com/in/divya-jha-20a65b317/recent-activity/all/

IX. GITHUB REPO



LINK : https://github.com/divyajha1711/Data-Analysis